

Ichhit Sapkota

AI/ML Research Intern

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Motivated and academically strong Computer Engineering final-year student with hands-on experience in AI/ML, computer vision, and data-driven systems. Actively involved in leadership, innovation, and community engagement through student clubs, startup competitions, and digital rights initiatives. Passionate about internet technologies, networking protocols, cybersecurity, and building ethical, secure, and scalable systems that create real-world impact.

EXPERIENCE

AI/ML Research Intern

Kathmandu

DeepQore Pvt Ltd

Jul, 2025 - Sept, 2025

- Worked on data-driven AI/ML models and real-world applications
- Gained exposure to secure system design, data handling, and model reliability
- Applied machine learning concepts to practical problem-solving scenarios

Secretary

Kathmandu

ICT Cosmos

Jun, 2025 - Present

- Organized technical workshops, peer-learning sessions, and student-led programs
- Promoted collaborative learning and technical discussions among students
- Actively involved in hackathons and technical training programs as a Trainer

Advisor

Kathmandu

Hult Cosmos

Nov, 2025 - Present

- Mentored students on startup ideation, innovation, and pitching
- Supported teams participating in national and international competitions

EDUCATION

Bachelor's in Computer Engineering

Kathmandu

Cosmos College of Management and Technology

Nov, 2022 - Present

Final Year Student

- Consistently ranked among top students with university topper positions in multiple semesters
- GPA details:
 - Semester I: 3.73
 - Semester II: 3.88
 - Semester III: 4.00
 - Semester IV: 4.00
 - Semester V: 3.96
 - Semester VI: 3.82
- Active involvement in technical leadership, innovation programs, and research-oriented projects

CAP-I

Kathmandu

Institute of Chartered Accountant Nepal

Jun, 2020 - Dec, 2020

Completed with 78% overall score

- Strong foundational knowledge in: Financial Accounting, Costing and Management Accounting, Taxation, Auditing, Business and Corporate Law

High School

Butwal

Glorious Secondary School

Jun, 2018 - Apr, 2020

College Topper

- Grade 11 GPA: 3.59

- Grade 12 GPA: 3.85

PROJECTS

VedaSync – AI Powered Classroom Management System

- Implemented real time CCTV-based facial recognition for attendance using Haarcascade, DNN (Caffe model), and Dlib ResNet with 68 landmarks and 128D facial embeddings
- Developed a cross-platform mobile application using Flutter
- Built backend services with Firebase Authentication and Firestore database
- Designed a Flask-based API for AI inference and facial recognition processing
- Integrated event and calendar management, analytics dashboard, and real-time notification features
- Focused on system scalability, accuracy, and data privacy considerations in smart classroom environments

Maternify – IoT Based Maternal Health Monitoring System

- Designed and developed an IoT-based maternal health monitoring system focused on improving prenatal care in areas with limited access to healthcare services
- Built a cross-platform mobile application using Flutter to display real-time health data
- Integrated live sensor data from ECG, EEG, pulse oximeter, blood pressure monitor, and other physiological sensors
- Implemented machine learning models to analyze physiological data and detect abnormal patterns
- Enabled real-time visualization of maternal and fetal health metrics within the mobile application
- Integrated smartwatch connectivity to continuously collect activity and health data from the mother
- Implemented fall detection and abnormal movement detection using accelerometer and gyroscope data
- Automated emergency alerts to predefined contacts or healthcare services in case of unusual activity or health anomalies
- Focused on data reliability, secure data transmission, and ethical handling of sensitive health information

ACHIEVEMENTS

Best Project Award

For “VedaSync – AI-Powered Classroom Management System” at Cosmos College of Management and Technology

Winner, Hult Prize On-Campus Competition – Maternify

Selected as the top team among six finalists, receiving the highest overall score from all judges

Semi-Finalist, National Hult Prize Competition – Maternify

Selected among the top teams nationwide, competing against 48 teams across the country

Finalist, Startup World Cup Nepal – Maternify

Selected among the top 10 teams nationwide after competing at the national level

TRAINING/CERTIFICATIONS

Python For Data Science

Covered data analysis, visualization, and machine learning fundamentals using Python

Microsoft Office

Practical training in Word, Excel, and PowerPoint for documentation, data handling, and presentations.

LaTeX Training

Hands-on training in technical document preparation, academic writing, and professional formatting using LaTeX

CS4ALL01: IT Fundamentals in Cyber Security (Intermediate & Higher Level)

Covered core cybersecurity concepts, system vulnerabilities, and foundational security practices.

CS4ALL02: Real-Time Cyber Threat Detection and Mitigation (Intermediate & Higher Level)

Focused on identifying, analyzing, and mitigating cyber threats in real-time environments.

CS4ALL03: Security Analysis and Risk Management (Intermediate & Higher Level)

Gained knowledge in risk assessment, security analysis, and cybersecurity decision-making frameworks.

CS4ALL04: Python for Cyber Security (Intermediate & Higher Level)

Applied Python for security automation, threat analysis, and cybersecurity-related scripting.

SKILLS

Python, Java, C, C++, Dart, Flutter, Machine Learning (SVM, KNN, Logistic Regression, Decision Trees, Deep Learning (CNN), Computer Vision, TensorFlow, Keras, PyTorch (basic), OpenCV, Dlib, Scikit-learn, NumPy, Pandas, IoT Systems, Sensor Data Processing (ECG, EEG, SpO2, BP), Accelerometer & Gyroscope, Data, Flask, Firebase (Authentication, Firestore), Computer Networking Fundamentals, Secure Data, Handling, Privacy-Aware System Design, Git, Linux (basic)

LANGUAGE

English, Nepali, Hindi

REFERENCES

Rabin Ghimire - *Project Manager*

DeepQore
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Ajaya Adhikari - *HOD, ICT*

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Chiranjibi Pandey - *DHoD, ICT*

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